Impacts of 48 years of consistent cropping, fertilization and land management on the physical quality of a clay loam soil

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Greenhouse and Processing Crops Research Centre, Agriculture and Agri-Food Canada, Harrow, Ontario, Canada N0R 1G0 (e-mail: dan.reynolds@agr.gc.ca). *Received 11 October 2013, accepted 3 March 2014. Published on the web 8 August 2014.*

Reynolds, W. D., Drury, C. F., Yang, X. M., Tan, C. S. and Yang, J. Y. 2014. **Impacts of 48 years of consistent cropping, fertilization and land management on the physical quality of a clay loam soil**. Can. J. Soil Sci. **94**: 403–419. Soil physical quality (SPQ) is often ignored or under-monitored in long-term field studies designed to determine the economic and environmental sustainability of agricultural practices. Accordingly, a suite of complementary soil physical and hydraulic parameters was measured using intact core samples to determine the SPQ of a Brookston clay loam under a long-term (48 yr) cropping, fertilization and land management study at Woodslee, Ontario. The SPQ under virgin woodlot, fertilized monoculture sod and unfertilized monoculture sod treatments was similar, with optimal SPQ occurring in the top 10–20 cm, but severely suboptimal SPQ occurring below 30 cm because bulk density (BD), relative field capacity (RFC) and saturated hydraulic conductivity (K_S) were excessive, and because organic carbon (OC), air capacity (AC) and plant-available water capacity (PAWC) were critically low. The SPQ for fertilized and unfertilized monoculture corn under fall moldboard plow tillage was similar and substantially suboptimal throughout the top 40–50 cm due to high or excessive BD and RFC, critically low OC, low or critically low AC and PAWC, and K_S that varied erratically from excessive to critically low. The SPQ under fertilized and unfertilized corn–oat–alfalfa–alfalfa rotations (corn and second-year alfalfa fall plowed) was similar and largely non-optimal below 10 cm, but largely optimal in the top 10 cm due to the ameliorating effects of numerous biopores and crop roots. A bimodal soil water release function fitted to release curve data showed that PAWC was determined by soil matrix porosity (P_M), and AC was determined by soil structure porosity (P_S). Strong inverse linear correlations between BD vs. P_M, BD vs. P_S and BD vs. OC provided site-specific estimates of optimal ranges and critical limits for PAWC, AC and OC, respectively. Although SPQ changed substantially among treatments, the changes did not extend below 30-to 40-cm depth, and were largely unaffected by long-term annual fertilization. The SPQ below 30- to 40-cm depth was similarly poor across all treatments, and is likely an inherent characteristic of the soil.

Key words: Soil physical quality, long-term field study, clay loam soil, agricultural sustainability, soil physical and hydraulic parameters

Reynolds, W. D., Drury, C. F., Yang, X. M., Tan, C. S. et Yang, J. Y. 2014. **Impact de 48 années de culture continue. Incidence de la fertilisation et de la gestion des terres sur la qualité physique d'un loam argileux**. Can. J. Soil Sci. **94**: 403–419. Les études sur le terrain de longue haleine conçues pour vérifier la pérennité économique et environnementale des pratiques agricoles tiennent rarement compte de la qualité physique du sol (QPS), ou ne surveillent pas suffisamment ce paramètre. Les auteurs ont mesuré une série de paramètres physiques et hydrauliques complémentaires du sol à partir de carottes intactes pour déterminer la QPS d'un loam argileux Brookston, dans le cadre d'une étude de 48 ans sur la culture, la fertilisation et la gestion des terres, à Woodslee, en Ontario. La QPS est la même pour la terre vierge du boisé, l'herbe sous la monoculture bonifiée et l'herbe sous la monoculture non fertilisée, et atteint sa valeur optimale dans la couche supérieure de 10 à 20 cm du sol. À plus de 30 cm de profondeur cependant, la QPS atteint une valeur considérablement inférieure à la valeur optimale, car la masse volumique apparente (MVA), la capacité relative au champ (CRC) et la conductivité hydraulique à saturation (K_S) sont trop élevées, alors que la concentration de carbone organique (CO), la capacité d'aération (CA) et le volume d'eau disponible pour la plante (VEDP) atteignent un niveau excessivement bas.

Abbreviations: **AC**, air capacity; **BD**, bulk density; $\mathbf{d_M}$, matrix domain modal equivalent pore diameter; $\mathbf{d_S}$, structure domain modal equivalent pore diameter; **-F**, annual chemical fertilization (modifier of MC, RC and Sod); $\mathbf{K_S}$, saturated hydraulic conductivity; **MC**, monoculture corn; **-NF**, no fertilization (modifier of MC, RC and Sod); $\mathbf{n_T}$, soil total porosity; **OC**, organic carbon content; **PAWC**, plant-available water capacity; $\mathbf{P_M}$, soil matrix domain porosity; $\mathbf{P_S}$, soil structure domain porosity; **PVD**, pore volume distribution; **RC**, corn-oat-alfalfa-alfalfa rotation; **RFC**, relative field capacity; **Sod**, monoculture bluegrass sod; **SPQ**, soil physical quality; **WL**, virgin woodlot area; $\boldsymbol{\theta_R}$, residual volumetric soil water content; $\boldsymbol{\theta_S}$, saturated volumetric soil water content; $\boldsymbol{\theta(\psi)}$, soil water release function

La QPS est similaire pour la monoculture de maïs avec et sans fertilisation après labour au versoir à l'automne, et s'avère sensiblement inférieure à la qualité optimale dans les 40 à 50 premiers centimètres du sol en raison d'une MVA et d'une CRC élevées ou excessives, d'une concentration de CO beaucoup trop faible, d'une CA et d'un VEDP faibles jusqu'à l'excès, et d'une K_S qui varie de façon erratique d'excessive à beaucoup trop basse. La QPS des assolements maïs-avoine-luzerne-luzerne avec ou sans fertilisation (labour automnal après le maïs et la deuxième année de luzerne) est similaire et s'avère largement inférieure à la valeur optimale passés dix centimètres, alors qu'elle est largement optimale dans les dix premiers centimètres du sol en raison de l'effet bénéfique de la profusion de pores biologiques et des racines de la culture. Une fonction bimodale représentant la libération de l'eau du sol ajustée avec les relevés de libération indique que le VEDP dépend de la porosité de la matrice du sol (P_M) tandis que la CA est déterminée par la porosité de la structure du sol (P_S). Une forte corrélation linéaire inverse entre la MVA et P_M, la MVA et P_S et la MVA et CO permet d'estimer la plage optimale et les limites du VEDP, de la CA et du CO pour le site, respectivement. Bien que la QPS change sensiblement avec le traitement, ces modifications ne s'observent pas à plus de 30–40 cm de profondeur et ne sont pas affectées par la fertilisation annuelle à long terme. À une profondeur supérieure à 30–40 cm, la QPS demeure piètre, peu importe le traitement, et caractéristique qui est sans doute inhérente au sol.

Mots clés: Wualité physique du sol, étude de longue haleine sur le terrain, loam argileux, pérennité de l'agriculture, paramètres physiques et hydrauliques du sol

It has been argued for decades that controlled, long-term field studies are the best means for assessing the true economic and environmental sustainability of agricultural practices (e.g., Stone et al. 1985; Jenkinson 1991; Janzen 1995; Robertson et al. 2008). This is because many important aspects of economic and environmental sustainability are not evident until years after an agricultural practice is instituted due to very slow rates of change for some properties and processes, poor "signal to noise" ratios, and complex parameter responses to exterior factors, such as weather. For example, Janzen (1995) and Drury and Tan (1995) had to employ strong smoothing techniques (5-yr running means) to detect important trends buried within several decades of noisy grain yield data; Jenkinson (1991) showed that soil organic carbon content was still changing slowly in long-term plots at Rothamsted, England after 150 yr; and Drury and Tan (1995) showed that some corn yields continued to change gradually in long-term field plots at Woodslee, Ontario after 35 yr.

A major component of agricultural sustainability includes maintenance of favourable soil physical and hydraulic properties (collectively known as "soil physical quality") in the root zone, as these properties exert a controlling influence on the soil's ability to imbibe, transport, store and supply crop-essential water, air, heat and dissolved nutrients (e.g., Topp et al. 1997). Despite this, few long-term field studies include systematic monitoring of more than a few (if any) soil physical quality parameters, focusing instead on other aspects such as fertility (N, P, K), trace elements, organic matter content, carbon sequestration, erosion, irrigation, pH, greenhouse gas emissions, crop yield, grain quality, etc. (e.g., Drury and Tan 1995; Janzen 1995, 2001; Mitchell et al. 1991; Drury et al. 1998). For example, out of 258 scientific publications generated between 1934 and 2012 from 13 crop rotation studies at six long-term field sites on the Canadian prairies (Campbell et al. 2012), only seven publications were dedicated to measuring soil physical quality and, of those, only two publications measured suites of physical quality parameters (i.e., Chang and Lindwall 1989; Pikul et al. 2006). The reasons for lack of soil physical quality measurement are not entirely clear, but seem to reflect lack of recognition of the importance of soil physical quality (at least initially), extensive labour and time required for collecting appropriate soil samples, and the need for specialized and dedicated laboratory infrastructure to conduct soil physical quality analyses.

A long-term cropping, fertilization and land management study has been in place at Woodslee, Ontario, Canada for over 48 yr. Like most long-term field studies, it was designed primarily to elucidate the impacts of consistent cropping and land management on yields and soil fertility, although it is also capable of monitoring the quantity and quality of tile drainage and surface runoff (e.g., Bolton et al. 1970). Dozens of scientific articles have been published from this study over the past 48 yr and, as might be expected, the publications deal predominantly with long-term trends in yields, fertility, organic carbon dynamics, greenhouse gas emissions, etc., with rather little attention paid to soil physical quality (e.g., Drury and Tan 1995; Drury et al. 1998, 2004). Three notable exceptions to this include McKeague et al. (1987), who described soil structure at the field site, Stone et al. (1985), who measured bulk density, porosity, wet aggregate stability and soil strength under some of the treatments, and Stone and Wires (1990), who measured bulk density, porosity and soil shrink-swell characteristics under some of the treatments. The objective of this work was therefore to characterize the physical quality of the Brookston clay loam soil under all treatments of this long-term field study by measuring a suite of complementary soil physical and hydraulic properties.

MATERIALS AND METHODS

Soil Physical Quality, Water Release Curve and Pore Volume Distribution Parameters

The soil physical quality (SPQ) parameters reported here include dry bulk density (BD; Mg m^{-3}), organic carbon content (OC; wt.%), air capacity (AC; $m^3\ m^{-3}$), plant-available water capacity (PAWC; $m^3\ m^{-3}$), relative field capacity (RFC; dimensionless), and saturated hydraulic conductivity (K_S; $m\ s^{-1}$). Also reported are parameters describing a bimodal soil water release function ($\theta(\psi)$; $m^3\ m^{-3}$), and the corresponding pore volume distribution (PVD; $m^3\ m^{-3}$). The first five parameters are well-established in the literature as indicators of soil physical quality (e.g., Reynolds et al. 2002, 2008), and so their working definitions, optimal ranges and critical limits are only briefly reviewed below for reader convenience. The bimodal $\theta(\psi)$ and PVD functions and associated parameters are less well known, however, and will therefore be described in greater detail.

The BD parameter is one indicator of mechanical resistance to root growth (e.g., Daddow and Warrington 1983), and is defined here by:

$$BD = \frac{M_S}{V_b} \tag{1}$$

where M_S (Mg) is mass of oven-dry soil and V_b (m^3) is bulk volume of undisturbed soil which is measured at pore water pressure head, $\psi = -1$ m, to account for possible shrink-swell behaviour (Reynolds et al. 2008). The optimal BD for root growth seems to be in the range 0.9 Mg $m^{-3} \leq BD \leq 1.2$ Mg m^{-3} regardless of soil texture (e.g., Olness et al. 1998; Reynolds et al. 2007). The BD upper critical limit (where root growth effectively stops) is strongly dependent on soil texture, ranging from about 1.35–1.40 Mg m^{-3} for heavy clays to about 1.50–1.60 Mg m^{-3} for loams to about 1.75–1.80 Mg m^{-3} for sands (Daddow and Warrington 1983). A BD upper critical limit of 1.47 Mg m^{-3} was assigned in this study, based on the contour plot in Fig. 3 of Daddow and Warrington (1983), and the average clay loam texture in the top 50 cm of the Brookston soil at the field site (i.e., 27 ± 6.6 wt.% sand; 34 ± 3.3 wt.% silt; 39 ± 5.3 wt.% clay – unpublished data).

The OC parameter is a measure of the total amount of organic material present in the soil, and it influences virtually all aspects of soil physical quality (Gregorich et al. 1997). The optimal OC range appears to be on the order of 3–5 wt.% for plant production (Craul 1999; Reynolds et al. 2009). For fine-textured soils, Greenland (1981) proposed an OC lower critical limit of ≈ 2.3 wt.%, below which tillage-induced loss of structure may occur, while Sojka and Upchurch (1999) speculated an OC upper critical limit of ≈ 6 wt.%, above which the soil may be overly prone to compaction.

Air capacity (AC) is an indicator of soil aeration, and may be defined as:

$$AC = \theta_S - \theta_{FC} \tag{2}$$

where θ_S ($m^3\ m^{-3}$) is saturated soil water content or soil porosity, and θ_{FC} ($m^3\ m^{-3}$) is the so-called field capacity water content, which is defined here as the equilibrium soil water content at "field capacity" pore water pressure head, $\psi_{FC} = -1$ m. It appears that AC > 0.12–0.17 $m^3\ m^{-3}$ is required for adequate aeration in fine-textured soils, while AC < 0.07–0.11 $m^3\ m^{-3}$ often results in periodic soil anaeorbiosis due to low gas diffusion rates combined with oxygen depletion by microbial activity (Carter 1988; Cockroft and Olsson 1997; Drewry 2006). In this study, AC = 0.14 $m^3\ m^{-3}$ was specified as the "lower optimal limit" for adequate soil aeration, while AC = 0.09 $m^3\ m^{-3}$ was specified as the "lower critical limit" below which periodic anaerobiosis would likely occur.

Plant-available water capacity (PAWC) is an indicator of the soil's ability to store crop-available water, and was defined here as:

$$PAWC = \theta_{FC} - \theta_{PWP} \tag{3}$$

where the permanent wilting point water content, θ_{PWP} ($m^3\ m^{-3}$), is the equilibrium soil water content at the "wilting point" pore water pressure head, $\psi_{PWP} = -150$ m. For medium- to fine- textured soils, it is considered that PAWC > 0.20 $m^3\ m^{-3}$ is "ideal"; $0.15 \leq$ PAWC ≤ 0.20 $m^3\ m^{-3}$ is "good"; 0.10 $m^3\ m^{-3} \leq$ PAWC ≤ 0.15 $m^3\ m^{-3}$ is "limited", and PAWC < 0.10 $m^3\ m^{-3}$ is "poor" with respect to root growth/function and drought resilience (Hall et al. 1977; Verdonck et al. 1983; Cockroft and Olsson 1997).

The so-called relative field capacity (RFC) partially combines the AC and PAWC indicators by expressing soil capacity to store air and water relative to θ_S or soil porosity (i.e., $\theta_S \approx$ soil porosity):

$$RFC = \left(\frac{\theta_{FC}}{\theta_S}\right) = \left[1 - \left(\frac{AC}{\theta_S}\right)\right] = \left(\frac{PAWC + \theta_{PWP}}{\theta_S}\right) \tag{4}$$

This parameter indicates which of the two "capacities" (i.e., AC or PAWC) is most limiting for a particular soil, and it has been proposed that the optimal range for RFC is 0.6–0.7, with RFC < 0.6 indicating "water limited" or "droughty" soil, and RFC > 0.7 indicating "aeration limited" soil (Linn and Doran 1984; Skopp et al. 1990; Olness et al. 1998; Reynolds et al. 2002).

The K_S parameter indicates the soil's ability to imbibe and transmit plant-available water to the crop root zone, as well as drain excess water out of the root zone (Topp et al. 1997). The parameter is defined by Darcy's law, which for flow through cylindrical cores takes the form:

$$K_S = \frac{QL}{\pi a^2 \Delta H} \tag{5}$$

where Q ($m^3\ s^{-1}$) is volumetric flow rate through the core, L (m) is core length, a (m) is core radius, and ΔH (m) is difference in hydraulic head across length, L. In humid climates, 5×10^{-6} m s$^{-1} \le K_S \le 5 \times 10^{-5}$ m s^{-1} is usually considered optimal for rapid infiltration and redistribution of crop-available water, minimal surface runoff and erosion, and prompt drainage of excess soil water (Reynolds et al. 2003). On the other hand, very rapid internal drainage associated with K_S values above the "upper critical limit", $K_S = 1.0 \times 10^{-4}$ m s^{-1} (e.g., Topp et al. 1997), can promote root zone droughtiness, while slow internal drainage associated with K_S values below the "lower critical limit", $K_S = 1.0 \times 10^{-6}$ m s^{-1} (e.g., Topp et al. 1997; McQueen and Shepard 2002), often promote inadequate root-zone aeration, reduced trafficability and excessive surface runoff and erosion.

The bimodal soil water release function, $\theta(\psi)$, and pore volume distribution function, PVD, apply for structured or "two-domain" soils, and characterize the volumes and size distributions of "matrix pores" and "structure pores". Matrix pores are defined here as the small voids between individual soil mineral particles; and structure pores are defined as the larger inter-pedal cracks, biopores (e.g., root channels, worm burrows, etc.), and inter-aggregate voids (Dexter 1988; Dexter et al. 2008). A convenient quasi-empirical $\theta(\psi)$ function for soils composed of "matrix pore" and "structure pore" domains may be derived from Dexter et al. (2008):

$$\theta(\psi) = \theta_R + P_M e^{\left(\frac{\psi}{-\psi_M}\right)} + P_S e^{\left(\frac{\psi}{-\psi_S}\right)} ; \psi \le 0 \qquad (6.1)$$

where θ_R (m^3 m^{-3}) is an effective "residual" soil water content (as $\psi \to -\infty$), P_M (m^3 m^{-3}) is effective soil matrix porosity, P_S (m^3 m^{-3}) is effective soil structure porosity, ψ_M (m) is the pore water pressure head at the "dry end" or "matrix" inflection point on the $\theta(\psi)$ curve, and ψ_S (m) is the pressure head at the "wet end" or "structure" inflection point on the $\theta(\psi)$ curve (Fig. 1a). Note also from Eq. 6.1 that when $\psi = 0$ (soil saturation):

$$\theta(\psi) = \theta_S \le n_T = \theta_R + P_M + P_S \qquad (6.2)$$

which shows that soil saturated water content, θ_S, equals (or nearly equals) soil total porosity, n_T (m^3 m^{-3}) (θ_S may be slightly less than n_T due to small amounts of entrapped air or encapsulated respiration gases), and that n_T is in turn equal to the linear sum of residual water content, θ_R, effective soil matrix porosity, P_M, and effective soil structure porosity, P_S.

The PVD may be defined as the semi-log slope of $\theta(\psi)$, plotted against equivalent pore diameter, d_E (μm), on a $\log_{10}$ scale (Jena and Gupta 2002), i.e.:

$$PVD \Rightarrow \frac{-d\theta(\psi)}{d\ln(-\psi)} \; vs. \; d_E; \psi < 0 \qquad (7)$$

where $\theta(\psi)$ is on a volumetric water content basis, and d_E is determined using the capillary rise equation (Or and Wraith 2002) in the form:

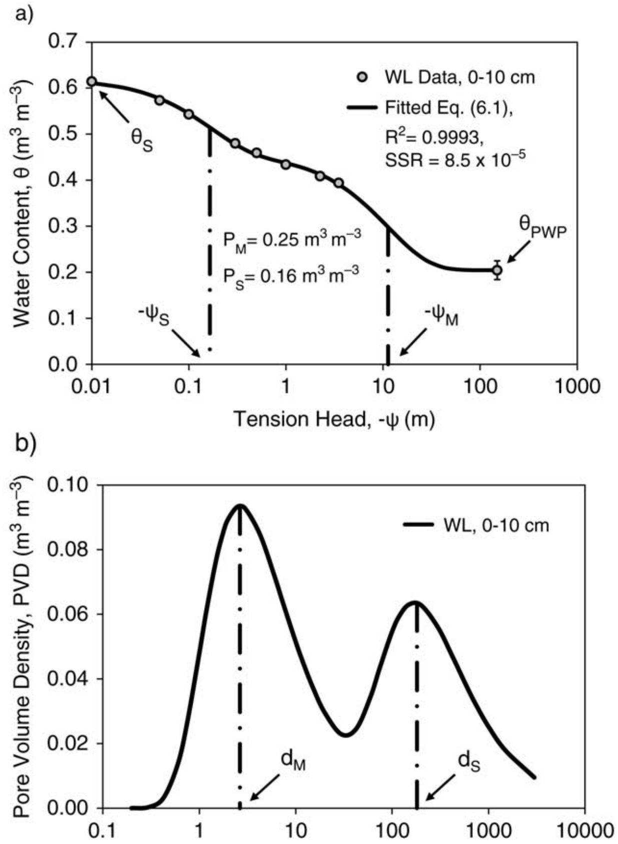

Fig. 1. Illustrative examples of: (a) bimodal soil water release function (Eq. 6.1) fitted to soil volumetric water content, θ, versus pore water tension head, $-\psi$, data for the 0- to 10-cm depth of virgin woodlot soil (WL); and (b) corresponding bimodal pore volume distribution, PVD, versus equivalent pore diameter, d_E. θ_S and θ_{PWP} are saturated and permanent wilting point volumetric soil water contents, respectively; P_M and P_S are soil matrix porosity and soil structure porosity, respectively; $-\psi_M$ is tension head at the matrix domain (dry end) inflection point and $-\psi_S$ is tension head at the structure domain (wet end) inflection point of the fitted soil water release function; d_M and d_S are modal equivalent pore diameter for the soil matrix domain and soil structure domain, respectively. The vertical "T" bars depict standard error ($n = 4$). Note that θ_S is measured at $-\psi = 0$, but plotted at $-\psi = 0.01$ m because of the $\log_{10}$ scale. Pore water tension head, $-\psi$, is the negative of pore water pressure head, ψ. Coefficient of determination (R^2) and sum of squared residuals (SSR) quantify the goodness of fit between Eq. 6.1 and the $\theta(-\psi)$ measurements.

$$d_E = \frac{4\gamma\cos(\omega)}{\rho_w g(-\psi)} \approx \frac{29.74}{-\psi} ; \psi < 0(\mathrm{m}); d_E(\mu\mathrm{m}); 20^{\circ}\mathrm{C} \qquad (8)$$

where $\gamma = 7.28 \times 10^{-2}$ N m^{-1} is pore water surface tension, $\rho_w = 998.2$ kg m^{-3} is water density, $g = 9.81$ m s^{-2} is gravitational acceleration, and $\omega \approx 0$ is the water-pore contact angle. Substituting Eq. 6.1 into Eq. 7 yields:

$$\frac{-d\theta(\psi)}{d\ln(-\psi)} = \psi\left[\frac{P_M}{-\psi_M}e^{\left(\frac{\psi}{-\psi_M}\right)} + \frac{P_S}{-\psi_S}e^{\left(\frac{\psi}{-\psi_S}\right)}\right]; \psi \leq 0 \quad (9)$$

which when plotted against Eq. 8 on a $\log_{10}$ scale produces a bi-modal PVD where:

$$d_M = \frac{4\gamma\cos(\omega)}{\rho_w g(-\psi_M)} \approx \frac{29.74}{-\psi_M}; \psi_M(\mathrm{m}); d_M(\mu\mathrm{m}) \quad (10a)$$

is the modal (most frequently occurring) equivalent pore diameter for the soil matrix domain, which drains at ψ_M, and

$$d_S = \frac{4\gamma\cos(\omega)}{\rho_w g(-\psi_S)} \approx \frac{29.74}{-\psi_S}; \psi_s(\mathrm{m}); d_s(\mu\mathrm{m}) \quad (10b)$$

is the modal equivalent pore diameter for the soil structure domain, which drains at ψ_S (Fig. 1b). Hence, Eqs. 6, 9 and 10 provide indicators of both the volume distributions and most frequently occurring pore sizes of the typically conjoined "matrix domain" and "structure domain" of a structured soil. Strictly speaking, Eqs. 6, 9 and 10 apply only for non-expanding (rigid) soils, which do not shrink or swell upon change in water content or pore water pressure head (i.e., n_T remains constant regardless of θ or ψ value). Work by Stirk (1954) and Nagpal et al. (1972) indicates, however, that the rigidity assumption remains essentially valid as long as $\psi \geq -150$ m, clay content ≤ 57 wt.%, and soil expansivity is moderate or less (e.g., free swelling index $< 100\%$, Sridharan and Prakash 2000). Note also that Eqs. 6, 9 and 10 are often expressed in terms of "tension head" (i.e. $-\psi$) rather than "pressure head" (ψ) to facilitate log transformation. Although the physical relevance, ideal ranges and critical limits for θ_R, P_M, P_S, d_M and d_S are still under development (e.g., Dexter et al. 2008), inclusion of these parameters along with the established SPQ indicators may provide important additional information and perspective on overall soil physical quality.

Field Site and Management

The study was situated on nearly level ($<0.5\%$ slope) Brookston clay loam soil (Canadian classification: Orthic Humic Gleysol; USDA classification: fine, loamy, mixed, mesic, Typic Argiaquoll) at the Hon. Eugene F. Whelan Research Farm, Agriculture and Agri-Food Canada, Woodslee, Ontario (lat. 42.2° N, long. 83.0°W). The average soil texture in the Ap horizon (0–20 cm) is 28.2 ± 4.0 wt.% sand, 34.4 ± 3.5 wt.% silt and 37.8 ± 4.5 wt.% clay, and the 45-yr average annual air temperature and precipitation are 8.9°C and 832 mm, respectively. The clay mineralogy in the Ap is predominantly illite, with small amounts of chlorite and vermiculite which impart mild to moderate expansivity (Stone and Wires 1990). The cropped area of the field site has been underdrained with 10-cm-diameter tile (average depth = 70 cm; spacing = 12.2 m) since 1955, but still has poor natural drainage (hydrological soil group, HSG-D) with frequent mottles throughout the Ap horizon and below (Richards et al. 1949; McKeague et al. 1987).

Six unreplicated cropping treatments were established in 1959, including:

- fertilized continuous (monoculture) corn (*Zea mays* L.) (MC-F)
- unfertilized continuous (monoculture) corn (MC-NF)
- fertilized monoculture bluegrass (*Poa pratensis* L.) sod (Sod-F)
- unfertilized monoculture bluegrass sod (Sod-NF)
- fertilized rotation corn (RC-F) in a corn–oat (*Avena sativa* L.)–alfalfa (*Medicago sativa* L.)–alfalfa rotation with all four phases present each year; first-year alfalfa established by spring under-seeding into oat phase.
- unfertilized rotation corn (RC-NF) in a corn–oat–alfalfa–alfalfa rotation with all four phases present each year.

The monoculture treatments plus each phase of both rotations were on 76.2-m-long by 9.2-m-wide plots, which were arranged side-by-side, separated by 3-m grassed berms, and longitudinally centered over one tile drain, which drained northward into an east–west drainage ditch (Fig. 2). Adjacent to the cropping treatments was an undrained virgin woodlot area of never cropped, fertilized or cultivated native deciduous trees and grasses (WL), which provided a "benchmark" for comparison purposes (Fig. 2). All fertilized treatments received a spring application of synthetic fertilizer (8–32–16), which supplied 16.8 kg N ha^{-1}, 67.2 kg P_2O_5 ha^{-1} and 33.6 kg K_2O ha^{-1} to the soil surface or seedbed, and fertilized corn received an additional side-dress application of 112 kg N ha^{-1} at the six-leaf stage. The unfertilized treatments and WL received no fertility amendments of any kind other than antecedent residues from the crop or native vegetation.

Cultivation included fall moldboard plowing (15- to 20-cm depth) of the monoculture corn and second-year alfalfa treatments, then secondary disking and harrowing in the following spring as required. Corn was planted in 1-m row widths at 37 000–62 000 seeds ha^{-1}, as dictated by available cultivars between 1959 and 2007. Weeds were controlled in the treatments and WL using common herbicides and application practices for the region. Harvesting included complete removal of corn grain in the fall (stover left on plots), plus cutting and removal (baling) of first-and second-year alfalfa one to three times per summer depending on growth. The monoculture sod treatments and native grasses in WL were mowed four to five times per summer, with cuttings left on the plots. Further detail on the field site and treatments can be found in Drury and Tan (1995) and Gregorich et al. (2001).

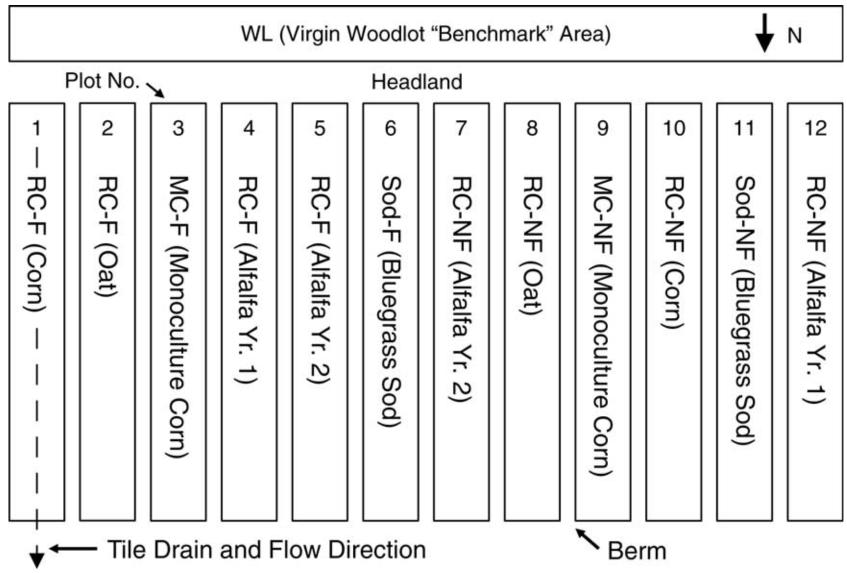

Fig. 2. Field plot layout and treatments at the start of measurements (Summer 2004). The cropped plots (numbered 1–12) were 76.2 m long by 9.2 m wide, separated by a 3-m raised berm, and centred longitudinally over a single tile drain (10-cm diameter, 70-cm average depth, 12.2-m spacing) that flows northward into an open east–west drainage ditch. The undrained "reference" or "benchmark" area (WL) was immediately south of a common headland (Headland) for the cropped plots, and supported native deciduous trees and grasses. The cropping treatments from left to right (in 2004) were: Plot 1, corn phase of chemically fertilized (-F) corn–oat–alfalfa–alfalfa rotation, RC-F (Corn); Plot 2, oat phase of fertilized rotation, RC-F (Oat); Plot 3, chemically fertilized monoculture corn, MC-F (Monoculture Corn); Plot 4, first-year alfalfa of fertilized rotation, RC-F (Alfalfa Yr. 1); Plot 5, second-year alfalfa of fertilized rotation, RC-F (Alfalfa Yr. 2); Plot 6, chemically fertilized bluegrass sod, Sod-F (Bluegrass Sod); Plot 7, second-year alfalfa of unfertilized (-NF) rotation, RC-NF (Alfalfa Yr. 2); Plot 8, oat phase of unfertilized rotation, RC-NF (Oat); Plot 9, unfertilized monoculture corn, MC-NF (Monoculture Corn); Plot 10, corn phase of unfertilized rotation, RC-NF (Corn); Plot 11, unfertilized monoculture sod, Sod-NF (Bluegrass Sod); and Plot 12, first-year alfalfa of unfertilized rotation, RC-NF (Alfalfa Yr. 1).

Sample Collection, Measurements and Calculations

Intact soil cores (10 cm diameter by 10 cm long) and companion grab samples were collected annually (using stainless steel sampling rings, manual drop hammer driver, and shovel) from five depths in each treatment during July–August 2004–2007 (4 yr). The top three sampling depths, i.e., 0–10, 10–20, and 20–30 cm ($n = 6$ per year), were collected at about 11-m intervals along the length of each cropping treatment plot. The remaining two sampling depths, i.e., 30–40, and 40–50 cm ($n = 3$ per year), were collected only from the southern plot ends and from the common headland area, respectively, to avoid undue plot disturbance (Fig. 2). Identical sampling times, depths and replication were used for WL. For the two rotations (RC-F, RC-NF), sampling was restricted to Plots 1 and 10 so that the same soil (Plot) and all four phases of the rotation (i.e., corn, oat, alfalfa, alfalfa) would be sampled over the 4-yr sampling period. Hence, sampling of the cropping treatments included Plots 1, 3, 6, 9, 10 and 11 (Fig. 2).

The intact cores were used to measure K_S, BD, θ_S, θ_{FC} and the soil water release curve, $\theta(\psi)$ (i.e., θ at $\psi = 0$, -0.05, -0.10, -0.30, -0.50, -1.00, -2.25, -3.50 m), using soil water flow and retention procedures in Reynolds (2008), Hao et al. (2008) and Reynolds and Topp (2008), respectively. The grab samples were used to obtain OC and the permanent wilting point soil water content, θ_{PWP} (m^3 m^{-3}) (i.e., θ at $\psi = -150$ m), using dry combustion (Skjemstad and Baldock 2008) and pressure plate (Reynolds and Topp 2008) techniques, respectively. The AC, PAWC and RFC parameters were calculated using the θ_S, θ_{FC} and θ_{PWP} measurements, and θ_{PWP} provided the final (lowest water content) point for the measured $\theta(\psi)$ curves.

Values for θ_R, P_M, P_S, ψ_M and ψ_S were obtained for each of the five depth ranges (i.e., 0–10, 10–20, 20–30, 30–40 and 40–50 cm) in each treatment by fitting Eq. 6.1 to the measured soil water release curves using the Solver® non-linear curve-fitting module (Frontline Systems, Incline Village, NV) in Microsoft Excel®. The corresponding bimodal pore volume distributions, d_M values, and d_S values were then determined using Eqs. 9, 10a and 10b, respectively.

In order to minimize random year-to-year variation (due to weather, etc.), the K_S, BD, OC and $\theta(\psi)$ data for WL, Sod-F, Sod-NF, MC-F and MC-NF were averaged over the 4-yr sampling period (i.e., 2004–2007) so that each measured value for each treatment and depth represents the mean of 12 to 24 individual measurements.

This was also done for the two four-crop rotations (i.e., RC-F and RC-NF) because the 4-yr average covers the latest of 12 complete rotation cycles during the past 48 yr, and should therefore provide a good representation of the average soil physical condition [i.e., K_S, BD, OC and $\theta(\psi)$ values] for each rotation. The bimodal $\theta(\psi)$ function (Eq. 6.1) was consequently fitted to the 4-yr averaged soil water release curve data for each treatment and depth, rather than to data for individual years (Fig. 1a). The goodness of fit between release curve data and Eq. 6.1 was quantified using coefficient of determination, R^2, and sum of squared residuals (SSR):

$$SSR = \sum_{i=1}^{n}[\Delta\theta(\psi_i)_i]^2;\ \mathrm{i} = 1, 2, 3, \ldots \mathrm{n};\ \mathrm{n} = 9 \qquad (11)$$

where $\Delta\theta(\psi_i)_i$ ($m^3\ m^{-3}$) is the difference between measured and equation-predicted soil water content at each of the nine pressure heads, ψ_i (m), comprising the measured soil water release curve.

Statistical Analyses

The residual maximum likelihood (REML) method (SAS Institute, Inc. 2012) was used to compare parameter means among depths within treatments and to the various optimal and critical limit values via the lsmeans procedure with the Tukey–Kramer adjustment for multiple comparisons ($P < 0.05$). Year, depth and subsample were used as random effects, and depth × treatment were used as fixed effects. Formal among-treatment statistical comparisons could not be conducted due to lack of treatment replication; however, among-treatment means with standard error bars which overlap or nearly overlap are likely not different, at least from a practical point of view. The K_S data were log-transformed before analysis due to approximate log-normal distributions, while the other parameters (BD, OC, AC, PAWC, RFC) were left untransformed due to approximate normal distributions.

RESULTS AND DISCUSSION

Soil Physical Quality Parameters

The SPQ indicator parameters (BD, OC, AC, PAWC, RFC, K_S) are given in Figs. 3–5, along with the relevant optimal ranges and critical limits. The generally small standard error bars for each parameter imply a high degree of spatial and temporal equilibrium for the seven treatments, which was not unexpected given the 48-yr duration of the experiment. The small standard error bars are also considered a justification for use of 4-yr parameter averages for all treatments including the two rotations (RC-F, RC-NF).

In virgin soil (WL), BD fell within the optimal range ($0.9\ Mg\ m^{-3} \leq BD \leq 1.2\ Mg\ m^{-3}$) in the top 20 cm, but exceeded the upper optimal limit ($1.2\ Mg\ m^{-3}$) at 20–30 cm ($P < 0.05$), was not significantly different from the upper critical limit ($1.47\ Mg\ m^{-3}$) at 30–40 cm, and exceeded the upper critical limit at 40–50 cm (Fig. 3a). The BD means initially increased significantly ($P < 0.05$) with increasing depth, then remained statistically constant for depths >30 cm (Fig. 3a). The corresponding OC, AC, and PAWC parameters were also optimal in the top 20 cm (except for OC at 0–10 cm, which was not significantly different from the OC upper critical limit), but rapidly fell below optimal at greater depths; i.e., OC was significantly below its lower critical limit at depths >20 cm, AC was not different from its lower critical limit at >20-cm depth, and PAWC was not different from its lower critical limit at >30-cm depth (Fig. 3b, c, d). The OC, AC and PAWC means all initially decreased significantly ($P <$ 0.05) with increasing depth, but OC became statistically constant below 30-cm depth, and AC and PAWC became statistically constant below 20-cm depth. The RFC parameter, on the other hand, was optimal only in the top 10 cm, and exceeded the upper optimal limit (0.7) by substantial and increasing amounts as depth increased, although significant differences ($P < 0.05$) did not occur at ≥20-cm depth (Fig. 3e). Perhaps surprisingly, K_S was highly uniform in magnitude and not significantly different ($P < 0.05$) throughout the top 40–50 cm, and it was also consistently equal to or significantly greater than its upper critical limit ($0.01\ cm\ s^{-1}$) (Fig. 3f). Based on these results (and notwithstanding excessive K_S), optimal soil physical quality conditions for plant growth in WL occurred only in about the top 10–20 cm, with substantially non-optimal conditions below 30–40 cm due to excessive BD and RFC, and critically low OC, AC and PAWC. It is also interesting to note that the WL soil was apparently both aeration-limited and potentially droughty at the same time, given that RFC >0.7 below 10-cm depth (Fig. 3e) and $K_S \geq 0.01\ cm\ s^{-1}$ throughout the entire measured profile (Fig. 3f). This might be explained by the fact that the soil under WL is clay loam with well-developed structure that graded downward from fine granular (0- to 5-cm depth) to medium granular (5- to 15-cm depth) to fine-medium subangular blocky (15- to 30-cm depth) to medium prismatic (30- to 75-cm depth) (McKeague et al. 1987). The fine clay loam texture evidently causes periodic aeration deficits in the soil matrix, as evidenced by the presence of grey–brown mottles (McKeague et al. 1987), and the well-developed structure produces highly water-conductive "structure pores" and thereby excessive permeability. In view of the above, virgin Brookston clay loam at the field site (WL) has inherently poor physical quality for plant growth below 30- to 40-cm depth, which was also suggested by McKeague et al. (1987).

With some exceptions, the BD, OC, AC, PAWC, RFC and K_S values for Sod-F and Sod-NF were well correlated with each other and with those of WL, i.e., they had generally the same significance patterns as WL with respect to depth, optimal ranges and critical limits, and the standard error bars of all three treatments frequently overlapped (Fig. 3a–f). Exceptions occurred primarily at 0- to 10-cm depth, where mean BD was

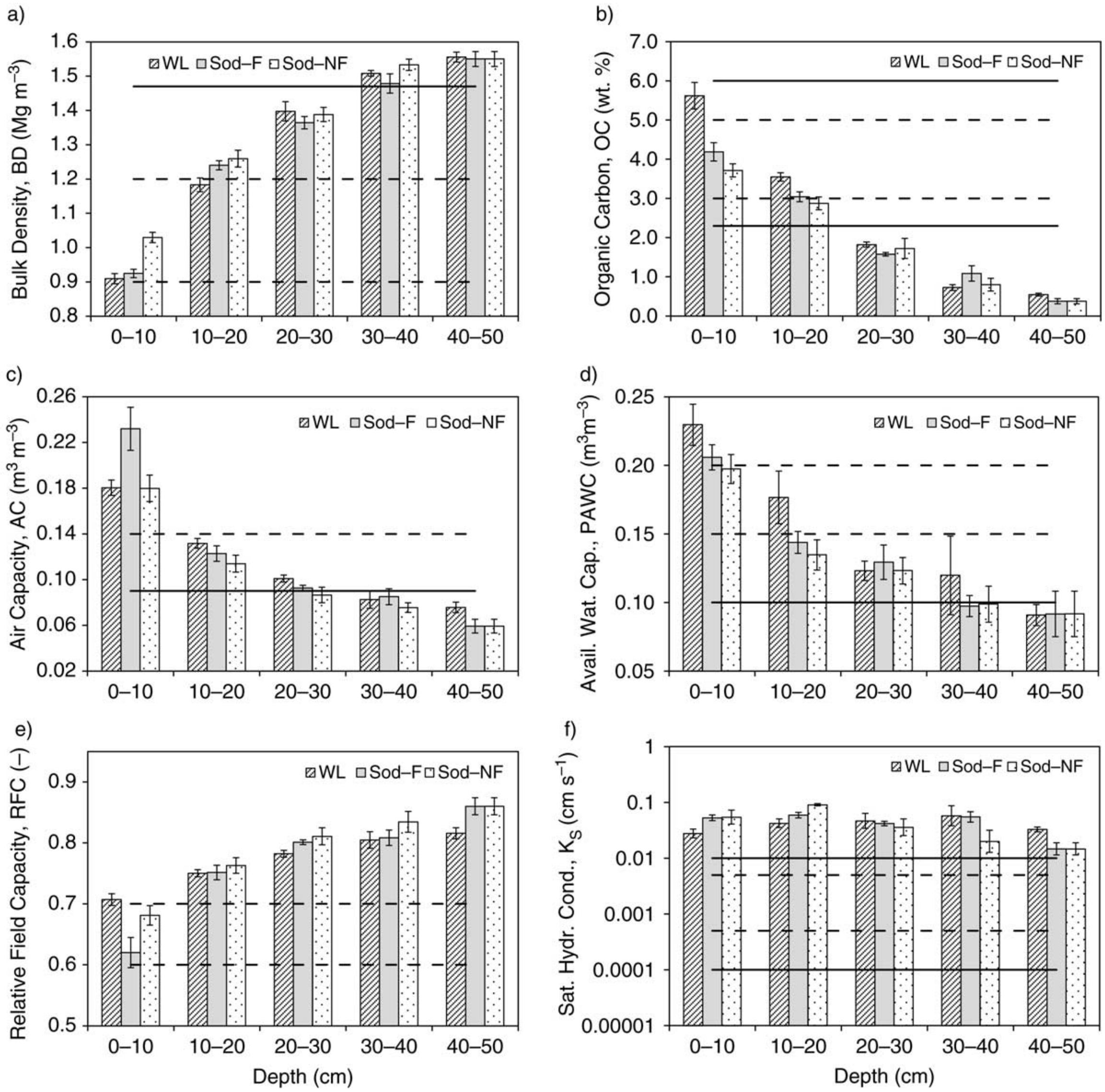

Fig. 3. Soil physical quality parameters versus depth for woodlot (WL), annually fertilized monoculture sod (Sod-F), and unfertilized monoculture sod (Sod-NF): (a) bulk density, BD; (b) organic carbon content, OC; (c) air capacity, AC; (d) plant-available water capacity, PAWC; (e) relative field capacity, RFC; and (f) saturated hydraulic conductivity, K_S. Horizontal dashed lines indicate parameter-specific upper/lower optimal limits; horizontal solid lines depict upper/lower critical limits (see text). The vertical ''T'' bars depict standard error ($n = 4$).

greater (but still optimal) under Sod-NF relative to Sod-F and WL (Fig. 3a); OC was lower (but still optimal) under both Sod-F and Sod-NF relative to WL (Fig. 3b); AC was greater under Sod-F than under Sod-NF and WL (Fig. 3c); PAWC was somewhat lower (but still optimal) under Sod-F and Sod-NF relative to WL (Fig. 3d); and RFC was lower (but still optimal) under Sod-F relative to Sod-NF and WL (Fig. 3e). The high degree of ''equivalence'' in soil physical quality among the three treatments is attributed to the fact that crop type and land management for Sod-F and Sod-NF (i.e., bluegrass crop with no soil disturbance and minimal trafficking) are reasonably similar to those of WL (i.e., native grasses with no soil disturbance and minimal trafficking), and this has in turn generated similar soil structure profiles (i.e., downward progression from granular to subangular blocky to prismatic). Note also that although long-term annual fertilization may have affected some SPQ parameters in the top 10 cm of bluegrass sod (i.e., BD, AC, RFC), no substantial or important differences between fertilized and unfertilized sod occurred below 10-cm depth (Fig. 3a–f).

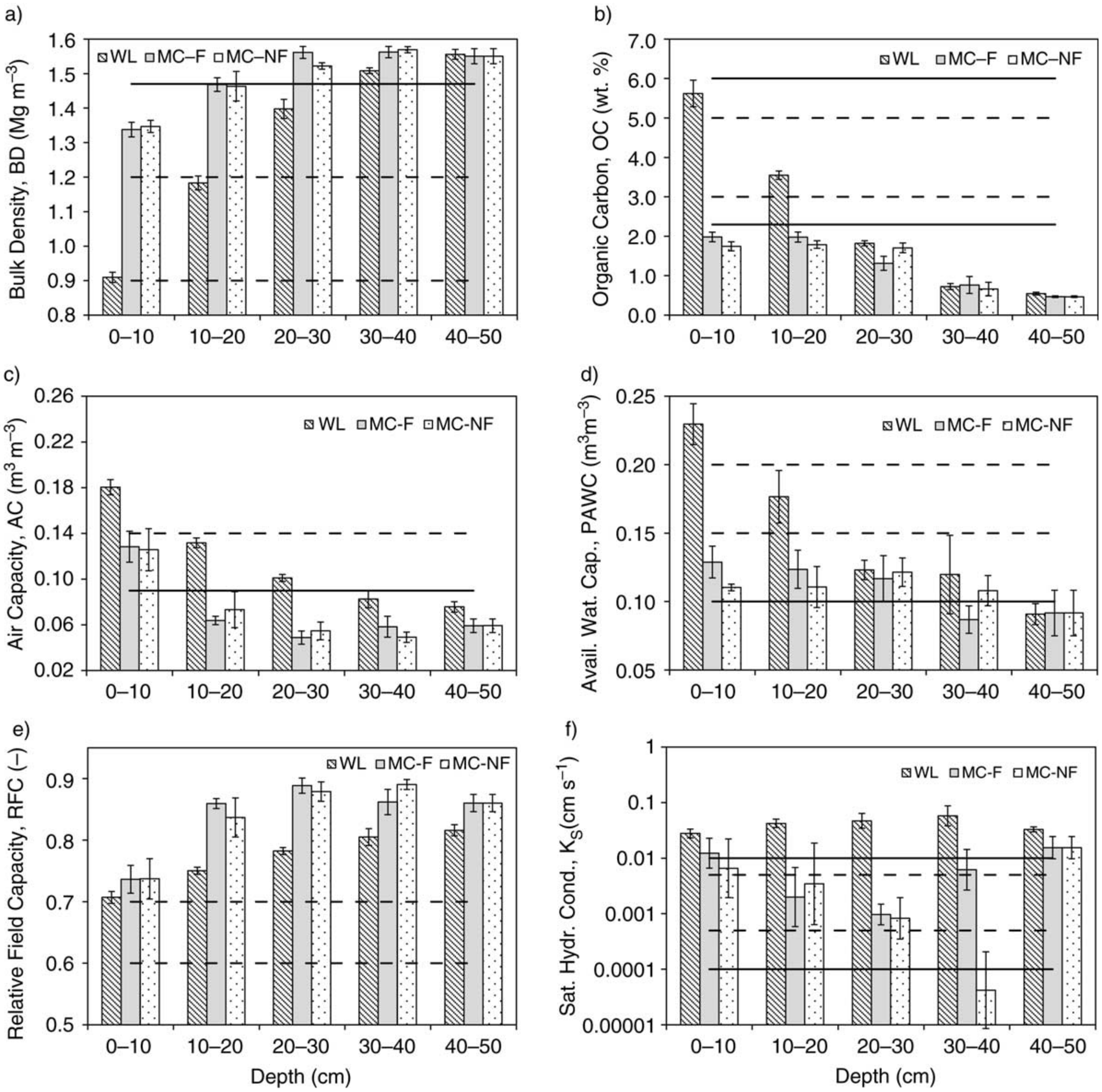

Fig. 4. Soil physical quality parameters versus depth for woodlot (WL), annually fertilized monoculture corn (MC-F), and unfertilized monoculture corn (MC-NF): (a) bulk density, BD; (b) organic carbon content, OC; (c) air capacity, AC; (d) plant-available water capacity, PAWC; (e) relative field capacity, RFC; and (f) saturated hydraulic conductivity, K_S. Horizontal dashed lines indicate parameter-specific upper/lower optimal limits; horizontal solid lines depict upper/lower critical limits (see text). The vertical "T" bars depict standard error ($n = 4$).

Long-term fertilized and unfertilized monoculture corn (MC-F and MC-NF, respectively) caused substantial changes in all SPQ parameters at most depths relative to WL. For both MC-F and MC-NF, BD was significantly above optimal at 0–10 cm, not different from the upper critical limit at 10–20 cm, and above the upper critical limit at depths greater than 20 cm (Fig. 4a); OC was significantly below the lower critical limit at all depths (Fig. 4b); AC was not different from the lower optimal limit at 0–10 cm, but significantly below the lower critical limit for all other depths (Fig. 4c); PAWC was significantly below the lower optimal limit at all depths, and not different from or below the lower critical limit in all cases but MC-F at 0–10 and 10–20 cm and MC-NF at 20–30 cm (Fig. 4d); and RFC was significantly above the upper optimal limit at all depths (Fig. 4e). In addition, K_S was highly variable, ranging from greater than or equal to the upper critical limit at 0–10 cm (MC-F) and 40–50 cm (MC-F, MC-NF), to not different from the lower critical limit at 30–40 cm (MC-NF) (Fig. 4f). For both treatments, the SPQ parameters at 0- to 10-cm depth tended to be significantly different ($P < 0.05$) from their respective values below 10 cm, except for OC and PAWC, which were not significantly different

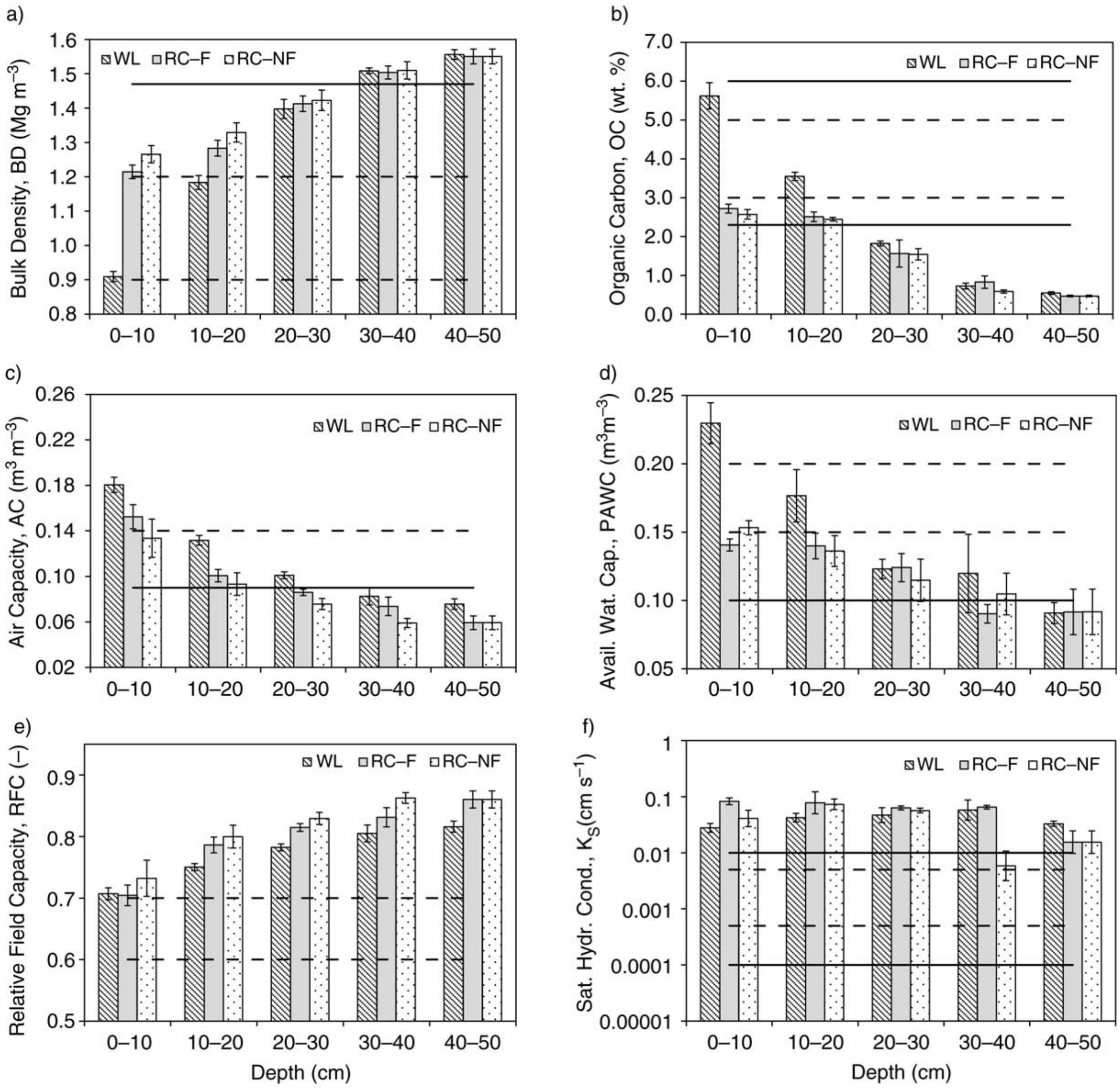

Fig. 5. Soil physical quality parameters versus depth for woodlot (WL), annually fertilized corn–oat–alfalfa–alfalfa rotation (RC-F), and unfertilized corn–oat–alfalfa–alfalfa rotation (RC-NF): (a) bulk density, BD; (b) organic carbon content, OC; (c) air capacity, AC; (d) plant-available water capacity, PAWC; (e) relative field capacity, RFC; and (f) saturated hydraulic conductivity, K_S. Horizontal dashed lines indicate parameter-specific upper/lower optimal limits; horizontal solid lines depict upper/lower critical limits (see text). The vertical "T" bars depict standard error ($n = 4$).

throughout the top 30 cm (Fig. 4b and d), and K_S, which differed erratically throughout the entire measured profile (Fig. 4f). These SPQ results are consistent with the prominent grey–brown mottling and largely massive, coarse blocky and coarse prismatic soil structure in the top 40–50 cm of MC-F and MC-NF (McKeague et al. 1987), especially the extreme variability of K_S (Fig. 4f), which implies that flow is primarily through isolated cracks and biopores surrounding dense, low-permeability soil clods. The SPQ results are also in line with previously observed corn rooting depth, which is limited primarily to the top 40 cm under MC-F and to the top 20 cm under MC-NF (Stone et al. 1987), i.e., root concentrations and depth distributions were likely impeded by the combined effects of extreme soil strength (excessive BD) below 20-cm depth (Fig. 4a) and insufficient soil aeration (critically low AC) below 10-cm depth (Fig. 4c). Note as well that long-term annual fertilization under MC-F increased corn root concentrations substantially in the top 30–40 cm relative to MC-NF (Stone et al. 1987), but had no appreciable effect on SPQ parameter values, except perhaps for K_S at 30- to 40-cm depth, which was not different from the upper critical limit under MC-F and not different from the lower critical limit under MC-NF (Fig. 4f). Hence, soil physical quality conditions for crop growth under MC-F and MC-NF were substantially non-optimal at all depths due to high or excessive BD and RFC, critically low OC, and low or critically low AC and PAWC.

The fertilized and unfertilized corn–oat–alfalfa–alfalfa rotations (RC-F and RC-NF, respectively) produced SPQ parameter values and significant difference patterns, which were largely intermediate between those of WL, Sod-F and Sod-NF on the one hand, and MC-F and MC-NF on the other (Fig. 5a–f). Specifically, BD equalled or exceeded the upper optimal limit at 0–10 cm, and equalled or exceeded the upper critical limit at depths greater than 30 cm (Fig. 5a); OC was not different from its optimal lower limit (RC-F) or critical lower limit (RC-NF) at 0–10 cm, but matched or fell below the critical lower limit at greater depths (Fig. 5b); AC equalled its optimal lower limit at 0–10 cm and matched or fell below the critical lower limit at greater depths (Fig. 5c); PAWC was not different from its optimal lower limit in the top 20 cm, but matched the critical lower limit at ≥ 30 cm (Fig. 5d); RFC equalled or exceeded its optimal upper limit at all depths (Fig. 5e); and K_S equalled or exceeded its critical upper limit at all depths (Fig. 5f). The reason for this ''intermediate'' behaviour between woodlot/sod and monoculture corn evidently stems from soil structure that is similar in form to MC-F and MC-NF (i.e., massive to coarse blocky and prismatic), but also perforated extensively by biopores and crop roots (corn, alfalfa) down to 60- to 70-cm depth (McKeague et al. 1987; Stone et al. 1987). As with the monoculture corn treatments, long-term fertilization under RC-F increased corn root concentrations substantially in the top 40–50 cm relative to RC-NF (Stone et al. 1987), but did not cause important changes in SPQ parameters, except perhaps for K_S at 30- to 40-cm depth, which exceeded the critical upper limit under RC-F, but was not different from the optimal upper limit under RC-NF (Fig. 5f). Hence, soil physical quality conditions for crop growth under RC-F and RC-NF were optimal or near-optimal in the top 10 cm, but largely non-optimal below 10 cm.

When the SPQ results are viewed in aggregate, it is clear that all treatments and WL have similarly poor soil physical quality below about the 30- to 40-cm depth (Figs. 3–5). This is likely an inherent characteristic of Brookston clay loam at the field site, as soil structure below that depth is consistently massive to coarse blocky and prismatic (McKeague et al. 1987).

Soil Water Release Curve Parameters

As exemplified in Fig. 1a, the two-domain soil water release function (Eqs. 6.1, 6.2) provided consistently good fits to the release curve data, with $1.0631\times10^{-5} \leq$ SSR $\leq 1.3294\times10^{-4}$, $0.9971 \leq R^2 \leq 0.9998$, and $1.012 \leq (\theta_R + P_M + P_S)/\theta_S \leq 1.023$. The function was therefore deemed appropriate and useful for the study. The slight (but consistent) 1.2–2.3% underestimate of $(\theta_R + P_M + P_S)$ by θ_S may reflect minor inadequacy in Eq. 6.1, or perhaps systematic bias in how Eq. 6.1 was fitted to the $\theta(\psi)$ data. The 33% overestimate of $(\theta_R + P_M + P_S)$ by θ_S obtained by Dexter et al. (2008), on the other hand, likely resulted from neglect of θ for $-0.1\ \text{m} < \psi \leq 0$ in their $\theta(\psi)$ measurements, and/or overestimate of average soil particle density in their Eq. 15.

It was found that $1.000177 \leq \theta_R/\theta_{PWP} \leq 1.000335$, which indicates that the ''residual'' water content in Eq. 6.1, θ_R, was effectively equivalent to the measured permanent wilting point water content, θ_{PWP}, rather than the water content as $\psi \rightarrow -\infty$ (Dexter et al. 2008). According to Czyż and Dexter (2012), this can occur when hydraulic continuity among pores is broken, thereby causing the soil water to exist as isolated, disconnected pockets, rather than as a continuous film lining pore walls. In any case, θ_R in Eqs. 6.1 and 6.2 can be viewed, at least for the soil in this study, as the proportion of soil total porosity (n_T) that is not useful to plant roots; or alternatively, $(P_M + P_S)$ in Eqs. 6.1 and 6.2 can be viewed as the ''plant-available'' proportion of n_T (i.e. ''plant-available'' porosity, n_{PAP}) that transmits, stores and supplies the soil air and soil water required for root growth and function.

Significant and strong linear correlations were found between AC and P_S ($0.9943 \leq R^2 \leq 0.9998$; $P < 1.85 \times 10^{-4}$ for slope), and between PAWC and P_M ($0.8703 \leq R^2 \leq 0.9999$; $P < 0.0207$ for slope), with average AC/P_S = 1.1335 (range = 1.1031–1.1728) and average P_M/PAWC = 1.1488 (range = 1.1190–1.1643) (data not shown). Hence, the soil's capacity to store air (AC) was determined largely by the soil's structure porosity (P_S), and the soil's capacity to store plant-available water (PAWC) was determined largely by the soil's matrix porosity (P_M). Note also that the average underestimate of AC by P_S is about the same as the average overestimate of PAWC by P_M, which from the definitions of AC and PAWC (Eqs. 2 and 3, respectively), suggests that the field capacity water content, θ_{FC}, may be underestimated by about 13–15%. This may in turn suggest that the ''correct'' field capacity pore water pressure head (ψ_{FC}) for the Brookston clay loam at the field site might be larger (less negative) than the ''standard'' value of $\psi_{FC} = -1$ m; and that fitting Eq. 6.1 to release curve data may provide a way of deducing the most appropriate ψ_{FC} for a particular soil and/or field site. Investigation of these possibilities is underway in a separate study.

By definition, the relationship between n_T and BD for rigid to moderately expansive soils can be expressed as:

$$n_T = 1 - \frac{BD}{PD} \tag{12}$$

where PD (Mg m^{-3}) is average soil particle density. Using Eq. 6.2 and the fact that $\theta_R \approx \theta_{PWP}$ (see above), Eq. 12 can be written as:

$$n_T = (\theta_R + P_M + P_S) = -\frac{1}{PD}BD + 1 \tag{13a}$$

$$n_{PAP} = (P_M + P_S) = -\frac{1}{PD}BD + (1 - \theta_{PWP}) \quad (13b)$$

$$n_{PAP} = (\theta_S - \theta_{PWP}) = -\frac{1}{PD}BD + (1 - \theta_{PWP}) \quad (13c)$$

where plant-available porosity, $n_{PAP} = (P_M + P_S) = (PAWC + AC) = (\theta_S - \theta_{PWP})$. Hence, plots of n_{PAP} vs. BD and $(P_M + P_S)$ vs. BD should be linear with slope $= -1/PD$ and intercept $= (1 - \theta_{PWP})$, and this was indeed the case within experimental error. Specifically, n_{PAP} vs. BD (Eq. 13c, Fig. 6a) was strongly linear ($R^2 = 0.9717$) and predicted PD = 2.43 Mg m^{-3}, which was only 4.1% less than the mean PD value obtained from 16 independent measurements (PD = 2.53 ± 0.05 Mg m^{-3}) using the standard pycnometer method (Flint and Flint 2002). In addition, the y-axis intercept predicted $\theta_{PWP} = 0.2084$ m^3 m^{-3}, which was only 1.3% greater than the mean measured value for the 0- to 10-cm depth ($\theta_{PWP} = 0.2057 \pm 0.0228$ m^3 m^{-3}). The individual P_M vs. BD and P_S vs. BD relationships (Fig. 6b, c) were also strongly linear ($R^2 = 0.9006$–0.9217), and when added as per Eq. 13b, they predicted PD = 2.39 Mg m^{-3} and $\theta_{PWP} = 0.1915$ m^3 m^{-3}, which are somewhat less accurate but still within 5.5 and 6.9% of the measured PD and θ_{PWP} values, respectively. Dexter et al. (2008) obtained similar inverse linear relationships between porosity and bulk density for sandy loam and loamy sand soils (their Eqs. 17–20); and the resulting PD predictions (2.17 Mg $m^{-3} \leq PD \leq 2.46$ Mg m^{-3}) spanned ours, but were substantially less than PD = 2.65 Mg m^{-3} assumed in their porosity calculation (their Eq. 15).

The P_M vs. BD (Fig. 6b) and P_S vs. BD (Fig. 6c) relationships were not significantly different ($P < 0.05$) with respect to slope and intercept, indicating that both matrix porosity (P_M) and structure porosity (P_S) decreased at effectively the same rate as BD increased. This was surprising, as it is often assumed (and found) that structure porosity (e.g., inter-aggregate and inter-pedal pore space) is more ‘‘fragile’’ than matrix porosity and therefore decreases at a greater rate as BD increases, especially if the soil has undergone compaction due to trafficking, tillage, etc. (e.g., Richard et al. 2001; Kutilek et al. 2006). As an example, Dexter et al. (2008) found in the top 45 cm of sandy loam and loamy sand soils that structure porosity decreased significantly and substantially with increasing BD, while matrix porosity remained virtually constant (their Fig. 5 and Eqs. 17 and 19). The equivalence of our P_M vs. BD and P_S vs. BD relationships (Fig. 6b, c) is therefore not understood, although strong and highly significant inverse linear correlations between BD and OC for every treatment except MC-F and MC-NF suggests that OC may be an important causal factor (Fig. 7). Scott and Wood (1989) obtained a very similar relationship between BD and OC for a silt loam soil with poor internal drainage, and data in Ruehlmann and Körschens (2009) shows that BD vs. OC relationships similar to Fig. 7 are typical for

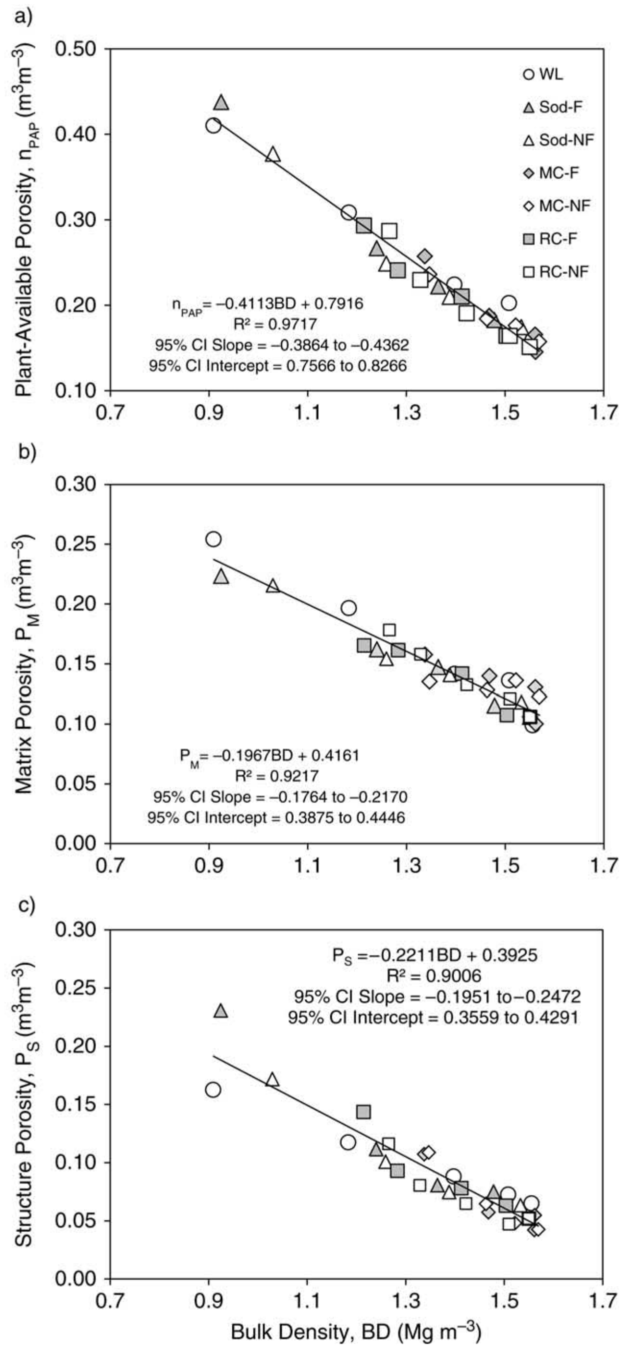

Fig. 6. Soil porosity versus bulk density (BD) for woodlot (WL), annually fertilized and unfertilized monoculture sod (Sod-F and Sod-NF, respectively), annually fertilized and unfertilized monoculture corn (MC-F and MC-NF, respectively), and annually fertilized and unfertilized corn–oat–alfalfa–alfalfa rotation (RC-F and RC-NF, respectively): (a) plant-available porosity, n_{PAP}; (b) matrix porosity, P_M; (c) structure porosity, P_S. Also included are n_{PAP} vs. BD, P_M vs. BD and P_S vs. BD regressions along with coefficients of determination (R^2) and 95% confidence intervals for regression slope (95% CI Slope) and regression intercept (95% CI Intercept).

the OC range in most mineral soils. It should also be remembered that organic matter has its own porosity and particle density, and is therefore expected to influence to varying degrees the soil's total porosity and average particle density.

The relationships in Figs. 6 and 7 allow optimal ranges and critical limits for n_{PAP}, P_M, P_S and OC to be estimated using the optimal range and critical upper limit for BD. For the BD optimal range (i.e., 0.9 Mg m^{-3} $\leq$ BD $\leq$1.2 Mg m^{-3}), the predicted optimal ranges are 0.30 m^3 m^{-3} $\leq n_{PAP} \leq$0.42 m^3 m^{-3}, 0.18 m^3 m^{-3} $\leq P_M \leq$0.24 m^3 m^{-3}, 0.13 m^3 m^{-3} $\leq P_S \leq$0.19 m^3 m^{-3}, and 3.1 wt.% $\leq$ OC $\leq$5.3 wt.%. For the BD critical upper limit (i.e., BD = 1.47 Mg m^{-3}), the corresponding critical lower limits are n_{PAP} = 0.187 m^3 m^{-3}, P_M = 0.13 m^3 m^{-3}, P_S = 0.068 m^3 m^{-3} and OC = 1.08 wt.%. Given that P_M and P_S correspond reasonably well with PAWC and AC, respectively, then for Brookston clay loam the optimal range and critical lower limit for PAWC might be estimated as 0.18–0.24 m^3 m^{-3} and 0.13 m^3 m^{-3}, respectively, while the optimal range and critical lower limit for AC are estimated as 0.13–0.19 m^3 m^{-3} and 0.07 m^3 m^{-3}, respectively. These predictions correspond surprisingly well with the generalized ranges and limits for fine-textured soils: i.e., the P_M optimal range of 0.18–0.24 m^3 m^{-3} overlaps the "good" and "ideal" ranges for PAWC; the P_M critical lower limit of 0.13 m^3 m^{-3} falls within the "limited" range for PAWC; the P_S optimal range of 0.13–0.19 m^3 m^{-3} includes the "lower limit" for adequate AC; the P_S critical lower limit of 0.07 m^3 m^{-3} falls within the "critical lower limit" range for AC; and the predicted OC optimal range of 3.1–5.3 wt.% matches well with the generalized OC range (3–5 wt.%). Hence, the n_{PAP} vs. BD, P_M vs. BD, P_S vs. BD and BD vs. OC relationships (Figs. 6 and 7), and the predicted optimal ranges and/or critical limits for n_{PAP}, P_M and P_S and OC, are plausible for Brookston clay loam soil at the field site, as well as good confirmations of the generalized ranges and limits in the literature.

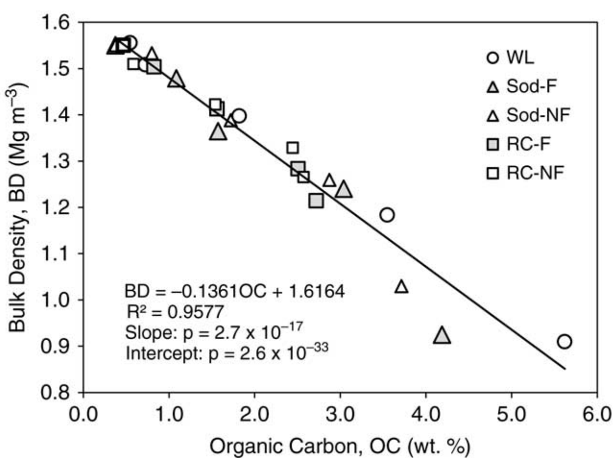

Fig. 7. Bulk density (BD) versus organic carbon content (OC) for woodlot (WL), annually fertilized and unfertilized monoculture sod (Sod-F and Sod-NF, respectively), and annually fertilized and unfertilized corn–oat–alfalfa–alfalfa rotation (RC-F and RC-NF, respectively). Also included is the overall BD vs. OC regression with coefficient of determination (R^2) and significance levels (*P*) for slope (Slope) and intercept (Intercept).

Using the regressions in Fig. 6, Brookston clay loam soil is predicted to lose all of its structure porosity (i.e., P_S = 0) at BD = 1.78 Mg m^{-3} (Fig. 6c), all of its plant-available porosity (i.e., $n_{PAP} = P_M + P_S = 0$) at BD = 1.92 Mg m^{-3} (Fig. 6a), and all of its matrix porosity (i.e., P_M = 0) at BD = 2.12 Mg m^{-3} (Fig. 6b). All of these values are well above the soil's estimated critical upper limit for root growth (BD = 1.47 Mg m^{-3}), which is probably due to the fact that root growth in fine-textured soil is controlled by pore size, aeration and several other factors besides mechanical resistance to root elongation (Daddow and Warrington 1983). Dexter et al. (2008), on the other hand, predicted BD = 1.58–1.89 Mg m^{-3} for zero structure porosity in selected sandy loam and loamy sand soils, which coincides well with the critical upper limit range (BD > 1.6 Mg m^{-3}) in Daddow and Warrington (1983) for coarse soil textures. It is also interesting to note that the BD vs. OC relationship in Fig. 7 predicts BD $\geq$ 1.30 Mg m^{-3} when OC falls below the critical lower limit where tillage may cause degradation of soil structure (OC $\leq$ 2.3 wt.%). This is consistent with our SPQ results in that only MC-F and MC-NF combine OC < 2.3 wt.%, BD > 1.3 Mg m^{-3}, and highly degraded soil structure (fragmented, massive, coarse blocky and prismatic) in the primary tillage zone (0–15 cm) (Fig. 4a, b).

As evidenced in Fig. 6, the soil water release curve parameters were not affected appreciably or systematically by long-term annual chemical fertilization versus no fertilization, as was also largely found for the soil physical quality indicator parameters (Figs. 3–5).

Pore Volume Distribution Parameters

The modal equivalent pore diameter for the soil matrix domain, d_M (Eq. 10a), ranged from 2.20 to 4.97 μm among all treatments and depths (overall mean and standard error of 3.46 μm and 0.098, respectively), while the corresponding pore diameters for the soil structure domain, d_S (Eq. 10b), ranged from 180.9 to 658.0 μm (overall mean and standard error of 382.6 μm and 21.4, respectively). This resulted in d_S/d_M = 54–209 depending on treatment and depth, indicating (as exemplified in Fig. 1b) that the soil matrix and structure domains were consistently well-defined and distinct in Brookston clay loam at the field site. The mean d_M and d_S values and d_S/d_M ratio obtained by Dexter et al. (2008), on the other hand, were substantially smaller (d_M = 1.55 μm, d_S = 48 μm, d_S/d_M = 31), indicating that their sandy loam and loamy sand soils had both smaller pores and less distinction between matrix and structure domains relative to our clay loam soil. This probably occurred because coarse soils typically have fewer and smaller aggregates, cracks, etc. than fine soils (e.g., White 2006). In addition, the Dexter et al. (2008) soils had both low mean OC ($\approx$1.2 wt.%) and high mean BD ($\approx$1.57 Mg m^{-3}), which would further encourage small pores and reduced aggregation. In any case, the d_M values obtained in this study and by Dexter et al. (2008) fit into the ‘‘available water’’ pore size category (0.2–30 μm) of White (2006) (their Table 4.2), while the d_S values fit mainly into the ‘‘drainage and aeration’’ size category (30–500 μm) with some overlap (in our case) into the ‘‘rapid drainage and macropore’’ category (500–5000 μm). This supports the above finding that plant-available water capacity (PAWC) in our soil was determined largely by matrix porosity (P_M), and air capacity (AC) was determined largely by structure porosity (P_S).

The d_M vs. P_M and d_S vs. P_S relationships varied substantially among treatments (Fig. 8) and must reflect treatment-induced changes in soil structure with changing porosity, given that soil texture at the field site is reasonably uniform, and soil porosity is determined by both pore size (as might be estimated by modal equivalent pore diameter) and pore number (e.g., Or and Wraith 2002). The systematic increases in d_M and d_S with decreasing P_M and P_S, respectively, under WL, Sod-F and Sod-NF (Fig. 8a, b) presumably reflects the observed gradation of soil structure under these treatments from fine-medium granular near the surface (where both P_M and P_S are greatest) to fine-medium subangular blocky and prismatic at depth (McKeague et al. 1987). In other words, fine-medium granular structure generates large numbers of small pores, while coarser blocky and prismatic structure produces smaller numbers of larger pores. In a similar fashion, the systematic increase in d_S with decreasing P_S under MC-F and MC-NF (Fig. 8d) is evidently caused by the downward coarsening of structure from fine fragmented and medium blocky/prismatic near the surface to coarse blocky/prismatic and massive at depth (McKeague et al. 1987). The soil matrix under MC-F and MC-NF, on the other hand, is largely massive with small numbers of irregularly spaced and isolated cracks and biopores (McKeague et al. 1987), and this is probably why the d_M values under these treatments tend to be large and unrelated to P_M (Fig. 8c). Given that the visible soil structure under RC-F and RC-NF is very similar to that of MC-F and MC-NF (McKeague et al. 1987), lack of relationship between modal equivalent pore diameter and porosity in both matrix and structure domains under RC-F and RC-NF (Fig. 8e, f) is likely due to the combined effects of degraded structure and proliferation of biopores and roots throughout the top 40–50 cm (McKeague et al. 1987; Stone et al. 1987).

Within each cropping system (i.e., monoculture sod, monoculture corn, corn–oat–alfalfa–alfalfa rotation), the d_M vs. P_M and d_S vs. P_S relationships were not affected systematically or appreciably by long-term annual chemical fertilization versus no fertilization (Fig. 8), as was also generally found for the soil physical

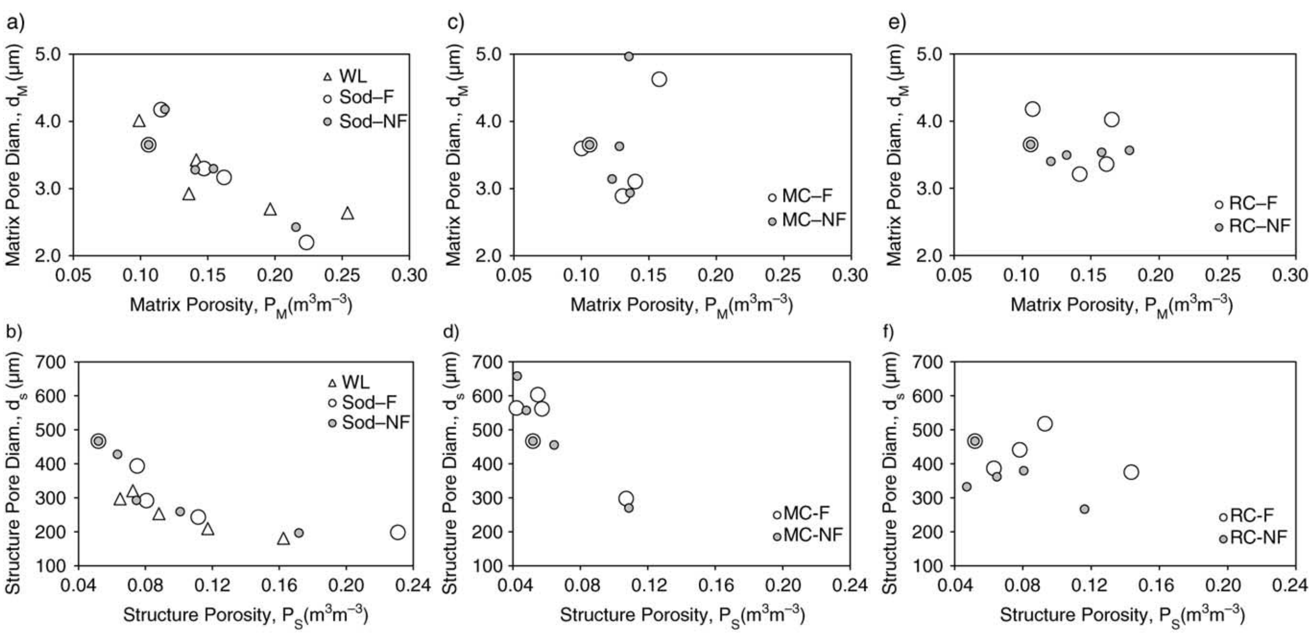

Fig. 8. Modal equivalent pore diameter versus porosity for woodlot (WL), annually fertilized and unfertilized monoculture sod (Sod-F and Sod-NF, respectively), annually fertilized and unfertilized monoculture corn (MC-F and MC-NF, respectively) and annually fertilized and unfertilized corn–oat–alfalfa–alfalfa rotation (RC-F and RC-NF, respectively). Panels (a), (c) and (e) depict matrix modal equivalent pore diameter (d_M) versus matrix porosity (P_M); panels (b), (d) and (f) depict structure modal equivalent pore diameter (d_S) versus structure porosity (P_S).

quality indicators (Figs. 3–5) and the water release curve parameters (Fig. 6).

CONCLUSIONS

Soil physical quality for plant growth under WL, Sod-F and Sod-NF was largely optimal in the top 10–20 cm, but substantially non-optimal below 30 cm because of excessive BD and RFC, and critically low OC, AC and PAWC. These treatments were also both aeration-limited and potentially droughty below about 10-cm depth due to RFC $>>0.7$ and excessive K_S, respectively. The overall "equivalence" in soil physical quality for WL, Sod-F and Sod-NF was attributed to similar plant types (i.e., continuous native grasses under WL vs. continuous bluegrass under Sod-F and Sod-NF) and similar land management (i.e., no soil disturbance and minimal trafficking), which in turn generated a soil structure profile in the top 50 cm that graded downward from fine-medium granular, to fine-medium subangular blocky, to medium prismatic. As a result, long-term (48 yr) continuous bluegrass sod production (Sod-F, Sod-NF) caused negligible degradation of soil physical quality in the primary root zone relative to virgin soil (WL).

The soil physical quality under MC-F and MC-NF was substantially non-optimal in the top 50 cm due to high or excessive BD and RFC, critically low OC, low or critically low AC and PAWC, K_S that varied erratically from excessive to critically low, and soil structure that was massive to coarse blocky/prismatic with prominent gray-brown mottles. The soil structure profile under RC-F and RC-NF was similar to that under MC-F and MC-NF, but soil physical quality was largely optimal in the top 10 cm and largely non-optimal below 10 cm, evidently because of the ameliorating effects of abundant biopores, and corn and alfalfa roots throughout the top 60- to 70-cm depth. Hence, long-term monoculture corn production (MC-F, MC-NF) caused extensive degradation of soil physical quality in the top 20–30 cm of the soil profile relative to virgin soil (WL), while long-term corn-oat-alfalfa-alfalfa rotation (RC-F, RC-NF) caused minimal to moderate degradation of soil physical quality in the top 10–20 cm.

A bimodal soil water release model (Eq. 6.1) similar to Dexter et al. (2008) provided excellent fits to soil water release curve measurements ($R^2 \geq 0.9971$, SSR $\leq 1.3 \times 10^{-4}$), and also gave estimates of soil residual water content (θ_R) matrix porosity (P_M) and structure porosity (P_S). Comparison to measurements showed that θ_R, P_M and P_S were highly correlated with θ_{PWP}, PAWC and AC, respectively, and that θ_R overestimated θ_{PWP} by $\leq$ 0.034%, P_M overestimated PAWC by an average of 15%, and P_S underestimated AC by an average of 13%. Hence, the volume of soil water not available to plant roots (θ_{PWP}) was determined by the model's residual water content (θ_R); capacity to store plant-available water (PAWC) was determined largely by soil matrix porosity (P_M); and capacity to store air (AC) was determined largely by soil structure porosity (P_S). The fact that the 15% average overestimate of PAWC by P_M was nearly the same as the 13% average underestimate of AC by P_S suggests that the actual field capacity soil water content (θ_{FC}) at the field site may be 13–15% larger than the value obtained using the "standard" field capacity pressure head ($\psi_{FC} = -1$ m).

Strong inverse linear correlations were found between bulk density (BD) and plant-available porosity (n_{PAP}), P_M and P_S ($R^2 \geq 0.9006$). These relationships were consistent with the definitional relationship between BD and porosity, and they estimated average soil particle density (PD) and θ_{PWP} within 4.1–5.5% and 1.3–6.9% of the measured values, respectively. Given that P_M and P_S estimated PAWC and AC, respectively, regressing BD against P_M and P_S provided plausible site-specific estimates of optimal ranges and critical limits for PAWC and AC. A strong inverse linear relationship was also found between BD and OC ($R^2 = 0.9577$), which predicted a site-specific OC optimal range (3.1–5.3 wt.%), as well as a site-specific BD limit (1.3 Mg m^{-3}) where soil structure at the field site may become susceptible to degradation by tillage.

Modal equivalent pore diameters obtained from the fitted pore volume distribution function (Eq. 9) compared well with pore size categories in the literature that are believed to control soil water storage, soil water drainage, and aeration. The structure domain modal pore diameters (d_S) fit into the high end of the "drainage and aeration" size category (30–500 μm), with some overlap into the low end of the "rapid drainage and macropore" category (500–5000 μm); and the matrix domain modal pore diameters (d_M) fit into the low end of the "plant-available water" pore size category (0.2–30 μm). These results are consistent with the finding that AC was determined largely by structure pores, and PAWC was determined largely by matrix pores. The d_M vs. P_M and d_S vs. P_S correlations ranged from strong to negligible, and appear to be controlled by type of soil structure.

Soil physical quality changed substantially among cropping system and WL, but the changes did not extend below 30- to 40-cm depth, and were largely unaffected by long-term annual chemical fertilization. Soil physical quality below 30- to 40-cm depth was similarly poor among cropping system and WL, and is likely an inherent characteristic of Brookston clay loam at the field site.

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